

HL7 FHIR: Technical Overview for Health Data Interoperability

What is FHIR? Origin and Purpose

Fast Healthcare Interoperability Resources (FHIR) is a standards framework created by Health Level Seven International (HL7) to enable the electronic exchange of healthcare information ¹. The HL7 community initiated FHIR around 2011 in response to growing demands for faster, easier data sharing as health records became digital ². FHIR's design builds on HL7's decades of experience with earlier standards (like HL7 v2, v3, and CDA), learning from their challenges and successes ³. The goal was to **simplify implementations without sacrificing information integrity** ⁴. Grahame Grieve (known as the "father of FHIR") and a small team spearheaded this effort after seeing firsthand how fragmented health data impeded care ⁵. In essence, FHIR was created as a modern solution to connect healthcare data across systems using web technologies, with a tight focus on implementability and real-world needs ⁶. Over the past decade, FHIR has evolved through multiple releases (from an initial 49 resource types to ~150 in the latest version) and has been widely adopted globally, benefitting providers, patients, and payers in thousands of applications ² ⁷. HL7 FHIR's overarching purpose is to provide a **flexible, open standard** that healthcare systems of all kinds can use to **interoperate** – sharing and understanding data efficiently to improve care quality, coordination, and innovation ⁸ ⁹.

FHIR Core Data Model and Architecture

At FHIR's core is a **resource-oriented data model**. A *Resource* is the fundamental building block of FHIR – a small, self-describing data packet that represents a specific healthcare concept (for example, *Patient*, *AllergyIntolerance*, *Medication*, *Observation*) ¹⁰. All exchangeable content in FHIR is defined as resources, and all resources share certain characteristics: they are defined in a consistent way, assembled from common reusable data types, include a standard set of metadata, and contain an optional human-readable narrative for clarity ¹¹. Each resource focuses on a coherent piece of data (often corresponding to a real-world entity or event in healthcare) and can stand alone. However, FHIR's design philosophy is **compositional** – complex use cases are addressed by linking or combining resources rather than by creating giant monolithic messages ¹². For example, a patient's record may be composed of a Patient resource linked to multiple related resources (Encounters, Observations, Procedures, etc.) via reference relationships. This modular **graph of resources** approach makes the data model very flexible and extensible, accommodating a wide range of healthcare scenarios ¹³ ¹⁴.

Architecture: FHIR adopts a resource-centric *RESTful API* architecture for most interactions ⁷. Systems implementing FHIR typically expose REST endpoints where each resource type has a base URL and each individual resource is identified by a unique URI. For instance, a server might expose `GET /Patient/123` to retrieve patient with ID 123. All the standard HTTP methods (GET, POST, PUT, DELETE, etc.) are used for CRUD operations on resources, leveraging the ubiquitous patterns of web APIs ⁹. This means developers can interact with healthcare data in a familiar way, similar to other modern web services. Under the hood, FHIR exchanges data primarily in **JSON or XML formats**, with support for both by default ¹⁵. These

representations are lightweight text formats that make it easy for disparate systems to parse and generate FHIR resources. (Notably, the JSON and XML formats, as well as the REST API usage, have reached normative status as of FHIR R4, underscoring their stability ¹⁶.) The use of web standards (HTTP, REST, OAuth for security – see SMART on FHIR later) means FHIR can be readily used in mobile apps, web applications, cloud services, and EHR integrations without special middleware ⁷. While RESTful exchange is the most common paradigm, the specification is not limited to REST – resources can also be used in other modes like messaging or document-based exchange when needed (e.g. bundling multiple resources into a clinical document) ¹⁷. This architectural flexibility lets FHIR accommodate both real-time transactional APIs and document-oriented workflows.

Interoperability and Conformance: A key design goal of FHIR is to enable **out-of-the-box interoperability** by keeping the base standard as straightforward as possible. The specification defines around 150 resource types that cover the “80%” of common use cases across clinical, administrative, and financial domains ¹⁸. Each resource has a broadly applicable definition so that most systems share a common core data structure. To address variability, FHIR includes a robust conformance framework. This is where **Profiles and Extensions** come into play. A *Profile* is a custom adaptation of a resource – defined using the FHIR **StructureDefinition** resource – that can constrain or extend the base resource for a specific use case or jurisdiction ¹⁹. For example, one can profile the *Patient* resource to require certain fields (like ethnicity in a US context) or restrict allowable values. An *Extension* is a mechanism to add a new data element to a resource that wasn't part of the base definition, in a controlled way that includes a globally unique URL to define the extension's meaning ²⁰. This built-in extensibility is vital: it ensures the base standard remains simple and implementable, while still allowing virtually any data to be represented when necessary by using extensions ²⁰. Many realm-specific requirements (e.g. a patient's race, national ID, or organ donor status) are handled via standard extensions or profiles rather than adding complexity to the core resource definitions ²¹. In practice, implementers create **Implementation Guides (IGs)** that specify how a set of resources, profiles, and extensions are used to solve a particular problem or realm (for example, the US Core IG defines how key resources must be structured for U.S. interoperability programs). FHIR also provides special metadata resources like **CapabilityStatement** (formerly *Conformance*) to describe a system's FHIR API capabilities, and **OperationDefinition** to define custom operations/APIs, which together enable automated negotiation and validation of interoperability between systems ²². All of these architectural elements – a consistent resource model, web-friendly RESTful API, flexible profiling, and explicit conformance artifacts – work in concert to make FHIR a **“platform” standard** on which diverse healthcare solutions can be built and still interoperate ²³. The specification emphasizes implementer needs by being concise and using familiar tech, so developers can rapidly build integrations without needing deep expertise in older HL7 paradigms ⁴. This focus on practicality and clarity is a major reason for FHIR's swift uptake in the healthcare IT community.

Integrating Clinical, Genomic, Behavioral, and Device Data

One of FHIR's strengths is its breadth – it supports integration of data from **multiple health domains** under one standard:

- **Clinical Data:** FHIR covers core clinical information traditionally found in EHRs and EMRs – such as patient demographics, problems/diagnoses, allergies, medications, immunizations, laboratory results, vital signs, imaging, procedures, care plans, appointments, referrals, etc. These are modeled with dedicated resources (e.g. *Patient*, *Condition*, *Medication*, *Observation*, *Procedure*, *CarePlan*, *Encounter*, *DiagnosticReport*, and many more). The specification is organized into modules like

Administration (for patient, practitioner, organization, **device**, etc.), *Clinical* (for observations, conditions, care plans, allergies, etc.), *Medications*, *Diagnostics*, *Scheduling*, *Financial*, and so on ²⁴ ¹⁸ – ensuring that most clinical workflows have an appropriate set of resources. By using FHIR, an application can seamlessly combine data about a patient's clinical status, their treatments and outcomes, and even billing or claims, since all share a common framework. The resources are also designed to reference each other (for example, an *Observation* resource for a lab result will reference the *Patient* resource, *Practitioner*, and possibly an *Encounter*), providing an integrated picture of clinical care.

- **Genomic Data:** FHIR has been extended to handle the rapidly growing area of genomics and precision medicine. HL7's Clinical Genomics work group has worked on defining FHIR representations for genomic information – for example, the **MolecularSequence** resource was introduced to represent DNA/RNA sequences, and more recently the **GenomicStudy** resource (in FHIR R5) to represent a genomic analysis involving one or more sequences. Through a combination of new resources and specialized profiles, FHIR can encode genetic test results, such as variant observations, genomic panels, and ancestry or pharmacogenomics data. In fact, HL7 published a Clinical Genomics **Domain Analysis Model** and has ongoing projects to ensure “genomics in FHIR” covers use cases like whole genome/exome sequencing, variant reporting, and genetic risk assessments ²⁵. For example, a genomic test result might be shared as a FHIR **DiagnosticReport** containing one or more **Observation** resources for genetic variants, or as a set of **MolecularSequence** resources with annotations. The flexibility of FHIR's data model (and its extension mechanism) allows even complex genomic data to be incorporated alongside clinical data. This enables integrated healthcare records that include both clinical observations and underlying genomic information when needed. The goal is that genomic results – say, identifying a tumor mutation or a pharmacogenomic allele – can travel with the patient's record in FHIR format to inform care anywhere. Standards groups like HL7 are actively publishing implementation guides to standardize these representations (e.g. the *FHIR Genomics Reporting IG*), and initiatives like **CodeX/GenomeX** (an HL7 FHIR Accelerator) are leveraging FHIR to improve cancer genomic data exchange ²⁶ ²⁷. In short, FHIR is being extended to natively support **precision medicine** data, so that clinical and genomic datasets interoperate on a single platform.

- **Behavioral and Social Data:** Beyond medical diagnoses and labs, FHIR is also used to represent behavioral health information, patient-reported outcomes, and social determinants of health. The *Patient* resource itself is designed to cover not just medical care but also contexts like psychiatric care, social services, and even personal fitness tracking ²⁸. For instance, behavioral health assessments or therapy notes could be shared using FHIR *Observation* or *DocumentReference* resources. Standard survey instruments (like depression screenings or substance use questionnaires) can be modeled with the *Questionnaire* and *QuestionnaireResponse* resources. There are profiles and projects (such as the **HL7 Gravity Project** for Social Determinants of Health) that use FHIR to capture data about housing, food insecurity, employment, and other social/behavioral factors via *Observations* and other resources. FHIR can handle **patient-generated health data** as well – e.g. data from wellness apps or patient devices tracking exercise, diet, mood, etc., can be represented as *Observations*. Because FHIR uses standard terminologies and coding where possible, behavioral data like mental health diagnoses (using DSM-5 or ICD codes) or assessment scores can be interoperably exchanged. The presence of workflow resources (like *CarePlan* or *Goal*) also supports behavioral health use cases where patient goals and interventions need to be tracked. In

summary, FHIR's resource set and extensibility make it capable of integrating **behavioral health and social context** data with the clinical record, facilitating a more holistic view of patient well-being.

- **Device and IoT Data:** FHIR includes resources to represent **medical devices** and the data they produce. The *Device* resource captures details about a piece of equipment or a tool (from an implantable pacemaker to a fitness wearable or a home glucose meter). There are related resources like *DeviceMetric* (for a device's measurements configuration) and *DeviceUsage/DeviceAssociation* (to record when a device is used for a patient). Readings from devices (vital signs, sensor measurements, etc.) are typically represented as FHIR Observations, which can carry device metadata and calibration details. By providing a standard format for device identifiers, manufacturer info, and linking devices to patients, FHIR enables easier **device data integration**. For example, a hospital's patient monitoring system could output vital signs in FHIR format, or a remote patient monitoring platform could share patient wearables data via a FHIR API. There are implementation guides mapping popular device data standards (such as IEEE 11073 for personal health devices) to FHIR, showing how to convert a Bluetooth blood pressure cuff's readings into FHIR Observations. In the FHIR workflow, devices are just another resource that can be referenced – e.g. an Observation has a `.device` element referencing the Device that generated it. Thus, a care team pulling a patient's record via FHIR can not only get clinical notes and labs, but also see data from **connected devices** (like continuous glucose monitoring results, smart inhaler usage, etc.) in the same standardized format. The FHIR *Administration* module explicitly includes devices as a core resource category ²⁹, underlining that devices are first-class citizens in the health data model. By unifying device data with EHR data, FHIR helps enable use cases like the Internet of Medical Things and telehealth, where disparate device streams need to feed into clinical workflows in an interoperable way.

Compliance, Identity Linkage, and Scalability

Standards Compliance and Security: As a formal HL7 standard, FHIR undergoes a rigorous balloting and versioning process to ensure **stability and compliance** with international standards norms. Notably, FHIR Release 4 (R4) marked the first time a large portion of the standard reached *normative* status (approved as an ANSI normative standard) ³⁰. This means key parts of FHIR (the base data types, JSON/XML formats, REST API, and important resources like Patient and Observation) are locked for backwards compatibility, giving implementers confidence that future changes won't break existing systems ³⁰. Many jurisdictions have incorporated FHIR into their interoperability regulations. For example, in the U.S., the 21st Century Cures Act and associated ONC rules mandate that certified EHRs expose a patient data API conforming to FHIR R4 (US Core profiles) and use the OAuth2-based SMART on FHIR security model. This effectively makes FHIR a compliance requirement for healthcare software in the US, and similar initiatives exist internationally. FHIR's design also includes resources and guidelines to support **privacy, security, and auditability**. The specification has a dedicated *Security and Privacy* module ³¹, and defines security-related resources such as **Consent** (to represent patient consent directives for data sharing) and **Provenance/AuditEvent** (to record data lineage and access events for compliance auditing). These allow organizations to meet privacy regulations (like HIPAA or GDPR) by using standard FHIR formats to track who accessed or contributed data ³¹. Furthermore, FHIR works with modern security protocols – it assumes transport-layer security (HTTPS) for all API calls, and the recommended authentication/authorization method is OAuth2/OIDC as profiled by SMART on FHIR (so that apps can securely obtain user-authorized access tokens). The HL7 FHIR community also publishes an implementer's **Security Best Practices** and has profiled FHIR for various security frameworks (e.g., US **HEART/UDAP** profiles for dynamic client registration and JWT-based

auth in networks). One prominent effort, the **FHIR At Scale Taskforce (FAST)**, has produced a set of security and scalability guidelines – including a Unified Data Access Profiles (UDAP) security IG – which is now being adopted as part of the USA's TEFCA nationwide health exchange framework ³². In short, FHIR provides the tools to build a secure, compliant ecosystem: from fine-grained consent and provenance records, to alignment with national trust frameworks like **TEFCA** that require common security and exchange protocols for FHIR-based data sharing ³². Organizations implementing FHIR can leverage these to ensure that data exchange not only meets interoperability needs but also regulatory and **trust** requirements.

Identity Linkage: In any health data infrastructure, matching and linking patient identities across systems is critical. FHIR offers multiple mechanisms to support **patient identity management** and record linkage. Each FHIR resource carries its own unique identifier (and can also include external identifiers, like medical record numbers, national IDs, etc., in a standard **Identifier** datatype structure) ³³ ³⁴. For linking duplicate records or identities, the *Patient* resource has a special element called `Patient.link`. The link allows one Patient resource to reference another, along with a relationship type (such as “replaces” or “refer” or “seealso”), indicating that they are the same individual or otherwise related ³⁴. In practice, if two Patient entries are found to be the same person, one can be linked to the other rather than merged blindly – preserving references and audit trails. As an HL7 blog succinctly describes: “*The link property in FHIR is used to assert that two (or more) patient resources are, in fact, the same person.*” ³⁵. In addition, FHIR includes a *Person* resource, which is designed to represent a single human across multiple roles or records – it can link to Patient, Practitioner, or RelatedPerson entries in different systems that actually refer to the same individual. This can be useful for building an enterprise Master Patient Index (MPI) or linking records from different facilities. The Person resource essentially provides a cross-reference hub, storing a person's demographic info and references to their various identifiers/records in other systems. By using Person and Patient.link, OZRYN could manage identity linkage: for example, if a patient shows up in two data sources under different IDs, those can be linked via one Person record or by linking one Patient to another as duplicates. Moreover, the FHIR community (in collaboration with IHE) has an established profile called **Patient Master Identity Registry (PMIR)** which outlines how to use FHIR for enterprise identity management, including handling merges, un-merges, and updates to patient identities. This means there is already guidance on how to use FHIR's identity linkage features in a scalable way. In summary, FHIR was built with the recognition that real-world data is messy – patients have multiple IDs and records – and it provides a standardized way to **link and reconcile identities** so that systems can maintain a unified patient view.

Scalability: As healthcare data exchange grows from individual transactions to population-level analytics, FHIR has evolved features to scale up. One major advancement is the **FHIR Bulk Data API** (also known as “Flat FHIR” or the Bulk Data Access IG). This specification defines a standardized way to export or transfer large sets of data (e.g. all patients in a population, or a full dataset from one system) in a single job, rather than one resource at a time. The bulk API uses an asynchronous pattern and newline-delimited JSON (NDJSON) files for efficient processing ³⁶. As HL7 describes, the Bulk Data Access standard “creates a common approach to handle data sharing at scale” – it allows systems to request **large volumes** of data (all relevant records for many patients) via a FHIR API and receive the results as a series of files that can be streamed or downloaded ³⁶. This is critical for use cases like research databases, public health reporting, or migrating data between systems. For example, a payer might use Bulk FHIR to pull a complete claims dataset for all members in a plan, or a hospital network might export an entire EHR dataset for analytics. The Bulk API ensures that even when dealing with millions of resources, the exchange is robust and standardized. Many EHR vendors and data platforms now support this Bulk FHIR approach, indicating that FHIR can handle big data workflows in addition to real-time APIs. Beyond bulk data, **performance and**

scalability are addressed by design choices like: resources being independent (allowing parallel processing and caching), support for search parameters and pagination on queries, and use of HTTP caching (ETags, Last-Modified headers) to reduce redundant transfers. There is also support for “**subscription**” and “**event**” paradigms (using webhooks or WebSockets) in FHIR, which allows scalable event-driven architectures (e.g. push notifications when new data arrives, rather than constant polling). Furthermore, the community-driven FAST initiative (FHIR at Scale Taskforce) has tackled challenges of scaling FHIR across networks – addressing issues like endpoint discovery, directory services for finding FHIR servers, and reliable **record location across multiple organizations**. The output of FAST, much of which is now being incorporated into HL7 specs and US policy, guides how to implement FHIR in a **nationwide, multi-network environment** with high performance and trust. For instance, FAST’s approach to a National Directory or to multi-party authentication is feeding into the US **TEFCA** (Trusted Exchange Framework and Common Agreement) so that **QHIN** networks can use FHIR reliably for cross-network queries ³⁷ ³⁸. The bottom line is that FHIR is not just for small-scale data sharing – it is deliberately being enhanced to **scale out** to very large systems and datasets, leveraging modern web scaling techniques. With bulk data, eventing, and careful use of infrastructure (cloud auto-scaling, etc.), a FHIR-based OZRYN data infrastructure could efficiently handle everything from an individual patient API call up to aggregating data for millions of patients. The rich tooling (many open-source FHIR servers and client libraries) and performance best-practices (like those from FAST) further ensure that FHIR solutions can be **scaled horizontally** to meet growing demand without sacrificing interoperability.

FHIR Versions, Profiles, and Ecosystem Maturity

Versioning (R4, R5 and beyond): The FHIR standard is iterative, with major “Release” versions that add capabilities while striving to maintain backwards compatibility. **FHIR Release 4 (R4)**, published at the end of 2018, was a milestone because it introduced the first normative content. This means parts of R4 are considered stable standards – for example, the Patient resource and several key APIs are normative ³⁰. R4 has been the foundation for many national programs and remains widely deployed. In fact, the U.S. and other regions have essentially standardized on FHIR R4 (v4.0.1) for current interoperability mandates. **FHIR Release 5 (R5)** was published in 2023 as the latest update. R5 is an STU (Standard for Trial Use) release that incorporates a broad set of enhancements: it adds dozens of new resource types, raises the maturity levels of many existing resources (some R4 resources that were trial use are more stable in R5), and introduces increased efficiencies in representations and workflows ³⁹. R5 features improvements for **enhanced interoperability, better data management, and streamlined workflows** according to HL7’s announcement ³⁹. Some notable changes in R5 include new resources for **Genomics (GenomicStudy)**, for **Public Health and Pandemic response**, and for **Clinical Reasoning/Measure reporting**, as well as refactoring in areas like **Medications** and **Evidence**. However, R5 is not yet broadly adopted in production – it serves as an intermediary step toward the next normative release. HL7 has signaled that **FHIR Release 6 (R6)** is on the horizon (likely in 2025 or 2026), intended to consolidate community feedback and possibly take more content normative. Due to timing and regulatory cycles, some realms are choosing to skip direct implementation of R5. For example, the HL7 US Realm Steering Committee decided that there will be no official US Core based on R5 – instead, the U.S. will stay on R4-based US Core until R6 is available as normative ⁴⁰. This decision was made to minimize disruption, since U.S. providers and vendors are currently deeply invested in R4 and the next big jump will be to an R6 that has widespread backing ⁴¹. This illustrates the careful balance in versioning: R4 is the stable workhorse, R5 is an incremental improvement for those who need its new features, and R6 will likely be the next broadly adopted base. The good news for implementers is that **backward compatibility** is a priority – systems on R4 can interoperate reasonably well with R5 systems, as most differences are minor (HL7 provides **comparison maps** and migration guides). For

OZRYN, this means one can confidently build on FHIR R4 today (which has extensive support and tooling), knowing that the path to R5/R6 is evolutionary. Each FHIR release is published with a full online specification, and HL7 maintains a *Directory of Published Versions* so that older specs remain accessible ⁴². This allows long-term systems to stick to a version or gradually upgrade as needed. In summary, FHIR's versioning approach provides **stability through normative content** and **innovation through trial use content**, ensuring the standard can grow without stranding existing implementations.

US Core and International Profiles: While the base FHIR spec is international, effective interoperability often requires further **profiling to regional or domain needs**. The **HL7 US Core** Implementation Guide (IG) is a prime example – it defines how FHIR is used for the United States interoperability context, aligning with the U.S. Core Data for Interoperability (USCDI) dataset required by federal rules. US Core constrains and extends certain FHIR R4 resources (Patient, AllergyIntolerance, Observation, etc.) to meet U.S. requirements and vocabularies. It is updated periodically to adopt new USCDI versions; for instance, US Core 6.1.0 was released to incorporate **USCDI v3** data elements into FHIR definitions ⁴³. This included additions like new profiles or extensions for *Sexual Orientation* or *Social Determinants*, matching the evolving U.S. standards. As regulations (like ONC's Cures Act Final Rule) have now set compliance timelines, the healthcare industry is transitioning from older US Core versions (e.g. 3.1.1 for USCDI v1) up to the latest (v6.x for USCDI v3/v4) ⁴⁴. Essentially, US Core serves as a **baseline profile** for any FHIR usage in the U.S. – EHR vendors and HIEs must support it, and applications are built to expect those conventions. Similarly, other countries and regions have their own core profiles: e.g. **UK Core**, **Canada Core**, **Australia Base**, **Netherlands Core**, etc., maintained by HL7 affiliates. These typically adapt FHIR to local code systems, identifiers, and clinical workflows.

Beyond national bases, there are cross-national or specialty profiles. A notable one is the **International Patient Summary (IPS)**. The IPS is an effort by HL7 (in collaboration with ISO, CEN, and IHE) to define a *portable snapshot* of essential health information about a patient for cross-border exchange ⁴⁵ ⁴⁶. HL7 published the IPS Implementation Guides in two flavors: one for CDA and one for FHIR (based on FHIR R4) ⁴⁷. The FHIR IPS IG defines a standardized composition (document) containing core data like allergies, medications, problems, immunizations, labs, etc., using globally agreed code systems (including a freely available subset of SNOMED CT for problem list terms) ⁴⁸ ⁴⁹. This IPS has even been adopted as an ISO standard (ISO 27269), underscoring its international importance. Countries across Europe and beyond are starting to pilot the FHIR-based IPS for exchanging patient summaries when citizens travel or move ⁵⁰ ⁵¹. Another example of international collaboration is the **Global Digital Health Partnership (GDHP)** and G7 nations endorsing FHIR IPS for interoperability across borders ⁵² ⁵³. For OZRYN, being aware of profiles like IPS means the project could align its data exports to global standards, facilitating broader compatibility. Besides IPS, HL7 has also introduced the **International Patient Access (IPA)** profile, which specifies a common set of FHIR API resources that any patient-facing app should be able to retrieve from any country's systems – a kind of international minimum baseline for patient data access ⁵⁴. This shows the FHIR ecosystem's maturity in addressing not just local but international interoperability challenges.

In addition, specialized communities (cardiology, oncology, public health, etc.) have created their own FHIR implementation guides. For example, **mCODE** (minimal Common Oncology Data Elements) is an HL7 FHIR IG for cancer data; **FHIR Clinical Guidelines** for computable care pathways; **Logica (formerly HSPEC) Care Maps**; and many others. These guides often build on base FHIR or national cores and provide domain-specific constraints. HL7 curates a registry of implementation guides (accessible via **fhir.org** and the HL7 registry) to help implementers find the right profiles for their needs ⁵⁵ ⁵⁶. The use of profiles is a double-edged sword: it offers precision for each use case, but also requires coordination to ensure different profiles

remain interoperable. That is why initiatives like HL7's *FHIR Accelerator Program* (which includes projects like Argonaut, Da Vinci, CARIN, Codex, Vulcan, Gravity, etc.) play a crucial role – they bring industry consortia together to agree on **standard profiles and APIs** for their domain (be it value-based care, payer-provider exchange, clinical research, etc.), rather than each organization creating its own variant ⁵⁷ ⁵⁸ . As a result, the FHIR ecosystem now has a rich library of IGs and profiles that greatly speed up implementation (developers can re-use these IG definitions and reference servers can be configured to them). OZRYN's architects should consider leveraging existing profiles (like US Core, IPS, and relevant domain IGs) as building blocks, to avoid reinventing data models and to maximize compatibility with external systems.

Ecosystem Maturity (SMART on FHIR, TEFCA, Tooling): The growth of FHIR has been accompanied by a robust ecosystem of tools, technologies, and initiatives that together signal its maturity. One key development is the **SMART on FHIR** framework. SMART (Substitutable Medical Apps, Reusable Technology) provides a standardized way to authorize and integrate third-party applications with clinical systems using FHIR. It specifies an OAuth2-based security layer and a launch protocol for apps, so that (for example) a patient-facing app or a clinician-facing app can securely connect to an EHR's FHIR API and retrieve data on behalf of a user. SMART on FHIR has essentially become the universal standard for API security in healthcare – it's mandated in the US ONC rules and is widely supported by EHR vendors. This means OZRYN can leverage SMART's approach to manage authentication/consent for data access. The presence of a common app platform also means there's a growing **marketplace of FHIR apps** and libraries that OZRYN could tap into or contribute to. Indeed, HL7 hosts an **App Gallery** and runs **FHIR App Roundtables** to showcase the innovative solutions being built on FHIR APIs ⁵⁹ ⁶⁰ . Everything from mobile personal health record apps, to clinical decision support modules, to research data capture tools are being enabled by FHIR and SMART. This demonstrates the **extensibility and reusability** of FHIR – new functionalities can be delivered as plug-in apps rather than monolithic EHR features, knowing that any SMART on FHIR-compliant data source can run them.

Another sign of maturity is the integration of FHIR into national infrastructure initiatives. We discussed TEFCA above – it's the US's nationwide health information exchange framework, and recent updates have made FHIR an integral part of its future. The **TEFCA FHIR Roadmap** lays out how networks (QHINs) will gradually incorporate FHIR-based exchange alongside older document-based exchange ⁶¹ . In 2024 and 2025, pilot projects (like the **HL7/FAST "Trebuchet" project** with Da Vinci and a QHIN) are already demonstrating how payer-provider data exchange can occur via "networked FHIR" at scale ⁶² ⁶³ . Regulatory drivers like CMS's policies on patient access and payer-to-payer data exchange also lean heavily on FHIR (e.g. the CARIN Blue Button IG for claims data, the PDex IG for payer data exchange). Moreover, governments worldwide (UK's NHS, Canada Health Infoway, Australia's Digital Health Agency, etc.) have endorsed FHIR in their strategic interoperability plans. This broad policy support indicates that FHIR is not a niche technology but rather a **cornerstone of health IT modernization**.

From a tooling perspective, the FHIR ecosystem offers abundant support. There are multiple open-source FHIR servers (such as **HAPI FHIR** in Java, **Smile CDR** (based on HAPI), **FHIR .NET API** by Microsoft, **IBM FHIR Server**, etc.), and many cloud providers have FHIR services (Microsoft Azure and Google Cloud provide managed FHIR store APIs, which are essentially ready-to-use FHIR servers). There are also dozens of client libraries in different languages, testing tools (the Touchstone testing platform, Postman collections for FHIR, etc.), and reference implementations provided by HL7 for things like the FHIR **Validator** (to check resource instances against profiles). HL7's continuous integration builds and **FHIR test Connectathons** (held several times each year) ensure that the standard is continuously verified by real implementations. In these Connectathons, developers from around the world convene (virtually or in person) to test

interoperability between their systems on chosen scenarios – including new draft features of FHIR – and this practice has been happening since FHIR’s early days ⁶⁴ ⁶⁵ . The result is a very **implementation-driven standard** with quick feedback loops. The connectathon culture means by the time a feature or IG is published, it’s already been proven out by multiple teams. For the OZRYN project, this culture and community support are invaluable: it can engage with the FHIR community to test its integrations and adopt best practices discovered by others. There are also **FHIR Accelerator** programs, as mentioned, where consortia focus on specific needs (OZRYN might intersect with **Vulcan** for research or **Gravity** for social health, for example, which are FHIR accelerator projects for those domains). All these factors – a strong community, active testing events, alignment with policy, a library of implementation guides, and readily available tooling – indicate that FHIR is a **mature and stable choice** for health data infrastructure in 2025 and beyond.

To illustrate the state of FHIR’s ecosystem: HL7 reports that FHIR is **already widely used in hundreds of production systems and apps worldwide** ⁶⁶ , and that number only grows as major technology companies and health IT vendors continue to invest in FHIR. The standard’s **R4 version is ANSI-approved and supported by government mandates**, ensuring longevity, while new releases (R5/R6) keep improving functionality driven by implementer needs. By leveraging FHIR, the OZRYN project would be building on a **modern, web-friendly, and globally recognized standard** that can unify clinical, genomic, behavioral, and device data. FHIR’s architecture will let OZRYN scale its data services and integrate with other systems (EHRs, labs, devices, national HIEs) relatively easily compared to older approaches. And by following the authoritative guides (HL7 FHIR spec, FHIR.org community resources, and HL7 profiles like US Core and IPS), OZRYN’s developers can implement with confidence that they are meeting interoperability norms. In summary, HL7 FHIR provides a **technically robust, community-vetted, and future-proof** foundation for health data interoperability – one that aligns perfectly with the needs of a comprehensive health data platform like OZRYN.

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1 3 4 8 10 11 12 14 17 18 19 20 22 24 29 31 42 Overview - FHIR v5.0.0

<https://www.hl7.org/fhir/overview.html>

2 5 6 FHIR Celebrates 10th Birthday

<https://info.hl7.org/fhir-10th-birthday>

7 9 15 16 30 66 HL7 Publishes FHIR® Release 4

<https://blog.hl7.org/hl7-publishes-fhir-release-4>

13 Home - SQL on FHIR v2.1.0-pre

<https://build.fhir.org/ig/FHIR/sql-on-fhir-v2/>

21 28 Patient - FHIR v5.0.0

<https://www.hl7.org/fhir/patient.html>

23 55 56 Implementation Guide Registry

<https://www.fhir.org/guides/registry/>

25 26 27 57 The Standard | The Official Blog of HL7 | clinical genomics

<https://blog.hl7.org/topic/clinical-genomics>

32 37 38 61 62 63 The Standard | The Official Blog of HL7 | TEFCA

<https://blog.hl7.org/topic/tefca>

33 34 35 Using FHIR to expose a Patient Identity Registry lookup service – part 1 | Hay on FHIR

<https://fhirblog.com/2013/10/20/using-fhir-to-expose-a-patient-identity-registry-lookup-service-part-1/>

36 58 Role of Tools like FHIR Bulk Data Access for Provider and Payer Data Exchange

<https://blog.hl7.org/leading-healthcare-stakeholders-commit-to-real-world-testing-of-hl7-fhir-bulk-data-implementation-guide-0>

39 HL7® Releases FHIR® v5.0 | Data Science at NIH

<https://datascience.nih.gov/content/hl7%C2%AE-releases-fhir%C2%AE-v50>

40 41 43 44 Dispatch From The US Realm: January WGM Recap & Important News About The Future of US Core & FHIR Release 5

<https://blog.hl7.org/dispatch-from-the-us-realm-january-wgm-recap-important-news-about-the-future-of-us-core-hl7-fhir-release-5>

45 46 47 48 49 50 51 52 53 54 Advancing the International Patient Summary

<https://blog.hl7.org/advancing-the-international-patient-summary-ips>

59 60 64 65 The Standard | The Official Blog of HL7 | SMART on FHIR

<https://blog.hl7.org/topic/smart-on-fhir>